

Artur Moreira de Carvalho

Junior Software Developer | Backend: PHP/Laravel, SQL/Oracle, Python | Accounting & Financial Systems

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SUMMARY

Junior software developer focused on backend software engineering: PHP 8/Laravel, legacy PHP, Oracle SQL and Python on a multi-tenant SaaS platform for accounting reconciliation. At Grupo PZM since 2024 (intern, promoted to junior in 2025), I turn reconciliation, period-close, tax and manual journal entry rules into code, write and fix Oracle queries, enforce authorization and tenant isolation, and back my changes with automated tests. I handle production support down to the root cause, use Python for automation and data validation, and work daily with AI coding agents.

SKILLS

- **Backend:** PHP 8 (OOP), Laravel, legacy PHP and modernization, REST APIs, business rules, multi-tenant systems, authorization and access control, layered architecture inspired by DDD; JavaScript, HTML, CSS
- **Data & databases:** SQL, Oracle (queries, views, sequences, migration scripts), relational databases, data validation and data quality, reporting and Excel exports
- **Automation:** Python (data validation and curation scripts, CLIs with typer and pydantic, developer tooling)
- **Quality & DevOps:** automated testing (PHPUnit/Pest, Cypress E2E in TypeScript, pytest), Git, GitHub Actions (public projects), Docker (local QA environment), Linux/WSL, production support, troubleshooting and root cause analysis, technical documentation
- **Applied AI:** AI-assisted development with coding agents (rules and context, local MCP server); external model as a semantic classifier with guardrails (prompt injection, data masking, deterministic fallback) and offline evaluation

EXPERIENCE

Junior Software Developer | Grupo PZM – PZM Enterprise | Aug 2025 – present | Belo Horizonte, Brazil

- Build and maintain the reconciliation, period-close scheduling, tax obligations and manual journal entry modules, implementing accounting business rules in PHP 8/Laravel and legacy PHP and fixing defects down to the root cause.
- Write and fix Oracle SQL for status, open-item and aging indicators and for tracing data discrepancies; for example, fixed a query that duplicated records in the tax module.
- Implemented re-authentication for critical actions with per-module control, plus division-level authorization for imported statement batches and an audit log of parameter changes, both with automated tests.
- AI Accounting Assistant for reconciliation support (in development, not deployed to production): designed the architecture and am the only developer on the module, working with AI coding agents. Deterministic core (rules, lexical retrieval, Oracle queries and templates); an external model only classifies intent and prompt-injection risk and breaks ties, on masked data, with timeout, circuit breaker and fallback. Offline evaluation with a held-out set and adversarial cases.
- Built a local QA environment with Docker, Python scripts for data curation and validation, and tooling for AI coding agents: rules and context, a read-only local MCP server and a code knowledge graph in SQLite.

Full Stack Development Intern | Grupo PZM – PZM Enterprise | Jan 2024 – Jul 2025 | Belo Horizonte, Brazil

- Shipped features and fixes in legacy PHP modules and Laravel services for accounting reconciliation, including division- and period-based flows, and supported user acceptance testing.
- Implemented duplicate-code detection for trial balance imports (CSV/TXT), flagging inconsistencies before consolidation.
- Standardized operational report extraction and Excel generation from Oracle queries, and maintained SQL scripts for data maintenance and technical investigations.
- Introduced Cypress to the project (configuration, commands and the first E2E tests for login and approval), a base the team has since extended, and started the PT/ES/EN internationalization.

PROJECTS

TrilhaDocs | Personal project, public code | Sep 2026 | github.com/ArturMoreiraDeCarvalho/trilhadoocs

- Problem: build accounting document packages (cover sheets and attachments) that can be checked later, without losing files or mixing up companies. Built, with AI coding agents, a local Python CLI: validate, preview, build and verify.
- Result: atomic, resumable ZIP packages with manifests and SHA-256 integrity checks; 100+ pytest tests and GitHub Actions CI on Ubuntu/Windows with Python 3.11/3.12. Stack: Python, pydantic, typer, pypdf, ruff.

Study projects (public code): Task Health API (Node.js REST API with OpenAPI, Docker, integration tests and CI) and Data Quality CLI (pre-load CSV validation in Python, pytest and CI).

EDUCATION

PUC Minas – Associate degree (Tecnólogo) in Systems Analysis and Development | Mar 2021 – expected 2026

Courses: CS50's Introduction to Artificial Intelligence with Python (edX); Alura: PHP, JavaScript ES6+, Docker, AWS; Rocketseat: Ignite ReactJS, Discover, NLW Node.js.

LANGUAGES

Portuguese (native) · English (intermediate) · Spanish (basic)